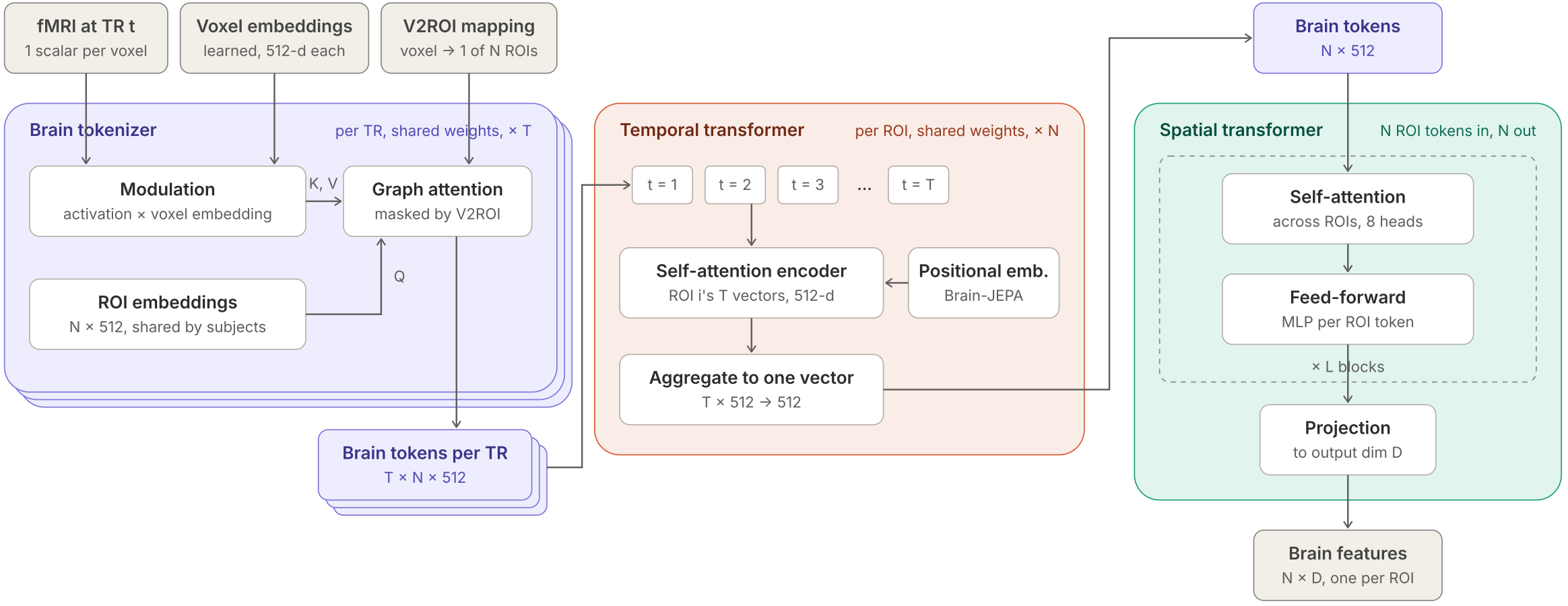
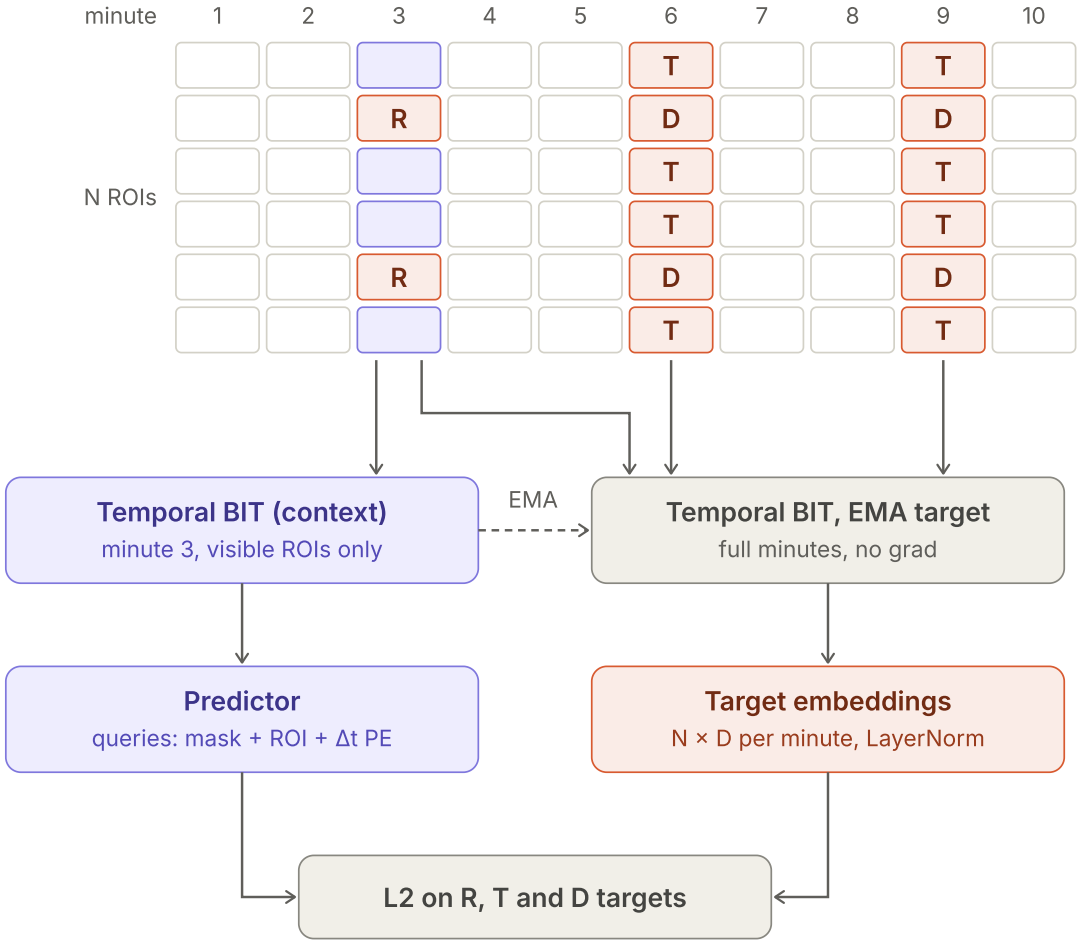
Temporal BIT: the encoder

Extending Brain-IT's BIT from a single TR to an fMRI time series of T TRs over N ROIs



Minute-level JEPA pretraining

Temporal BIT encodes one 60 s minute into N × D ROI embeddings; a 10-minute scan becomes an N ROIs × 10 minutes grid of tokens



Grid legend

- **Context**: visible ROIs of the context minute (here minute 3).
- **R** **Cross-ROI**: masked ROIs in the context minute. Same minute, other ROIs. Teaches inter-ROI coupling (functional connectivity).
- **T** **Cross-time**: the visible ROIs in other minutes. Same ROIs, other minute. Teaches stable, subject-level properties.
- **D** **Double-cross**: masked ROIs in other minutes. Neither ROI nor minute was seen; hardest target.

Training step

1. **Context encoder**: Temporal BIT on one minute; masked ROIs are removed entirely (voxels, temporal pass, brain token and query).
2. **Target encoder**: EMA copy of Temporal BIT on full minutes, no gradients; outputs N × D per minute.
3. **Predictor**: narrow BIT cross-transformer, one query per target = mask token + ROI embedding + minute offset Δt.
4. **Loss**: L2 between predicted and target embeddings over R, T and D. Context minute, masked ROIs and target minutes are resampled every step.

Downstream: keep the context encoder, discard the predictor. A scan yields one N × D embedding per minute, aggregated across minutes (mean or a small transformer) for diagnosis. Target types follow Brain-JEPA's spatiotemporal masking (Cross-ROI, Cross-Time, Double-Cross).